Monitoring Fractional Floating Algae Cover Over Eutrophic Lakes Using Multisensor Satellite Images: MODIS, VIIRS, GOCI, and OLCI

Kun Xue, Ronghua Ma, Zhigang Cao, *Member, IEEE*, Ming Shen, Minqi Hu, and Junfeng Xiong

***Abstract*— Coarse-resolution sensors have been used operationally to monitor floating algal blooms with a near daily revisit in coastal and inland waters. Most of the current methods in estimating fractional floating algae cover (FAC) were based on the linear pixel un-mixing assumption. In this study, a new FAC model following a logistic curve was developed and applied to multisensor satellite data in two large shallow eutrophic lakes, Lake Taihu and Lake Chaohu, in China. The results indicated that after resampling to 250 m, match-up pairs of Moderate Resolution Imaging Spectroradiometer (MODIS), The Visible and Infrared Imager/Radiometer Suite (VIIRS), Geostationary Ocean Color Imager (GOCI), and Ocean and Land Color Instrument (OLCI) possessed consistent Rayleigh corrected reflectance (R_{rc}) and floating algae index (FAI) or AFAI (alternative FAI). The FAC model was developed based on the simulated AFAI data of GOCI using point spread function (PSF) and bloom percent derived from Operational Land Imager (OLI) and then was applied to MODIS, VIIRS, and OLCI. Compared with the linear pixel un-mixing method, the FAC model reflects the asymptotic reflectance saturation in the near-infrared (NIR) band with the accumulation of blooms. Besides, the equivalent bloom area (EBA) of GOCI was validated using OLI-matched pairs with UPD 37.6% ($N = 39$, $R^2 = 0.96$). The spatial-temporal dataset of FAC (2002–2020) shows that Lake Taihu and Lake Chaohu experienced severe algal blooms after 2010, partly resulting from the higher frequency of multisensor data. This study provides a method for building a lasting and comparable FAC dataset using multisensors.**

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***Index Terms*— Equivalent bloom area (EBA), eutrophic lakes, fractional floating algae cover (FAC), multisensor, pixel un-mixing.**

I. Introduction

THE frequent occurrence of algal blooms in coastal and inland waters threatens drinking water, human health, water quality, and the ecological environment [1], [2]. Satellite remote sensing has been widely used to detect harmful algal blooms in oceanic and inland waters [3], [4]. Many large shallow lakes, such as Lake Taihu [5], [6], Lake Chaohu [7], Lake Winnipeg [8], [9], Lake Erie [10], and Lake Laguna [11], have suffered from eutrophication and algal bloom breakthroughs. Recently, global mapping using Landsat satellite series data revealed an increasing tendency of algal blooms in lakes and reservoirs [12], [13], in the most pronounced increases found in Asia [12], [14]. In particular, significant increases in bloom area, frequency, and duration of algal blooms were observed in 48 lakes and reservoirs in China from 1983 to 2017 [15].

Unlike other phytoplankton species, cyanobacteria can rapidly alter their buoyancy, and in calm weather, they can form surface or subsurface scums. The floating algal blooms of this type exhibit high reflectance in the near-infrared (NIR) band, similar to land surface vegetation, which leads to failure in accurate atmospheric correction and bio-optical inversion algorithms [16]. Therefore, current studies usually use the top-of-atmosphere (TOA) reflectance [13], surface reflectance [12], or Rayleigh corrected reflectance (R_{rc}) [16] of satellite data to detect bloom scums or floating algal bloom. The maximum chlorophyll index (MCI) [17], cyanobacteria index (CI) [18], and maximum peak height (MPH) [19], [20] were designed to quantify the chlorophyll-a fluorescence at approximately 681 nm. Several bloom detection indexes, for instance, floating algae index (FAI) [16], alternative FAI (AFAI) [6], virtual baseline floating macroalgae height (VB-FAH) [21], and index of floating green algae for GOCI (IGAG) [22], were developed to quantify the red-edge reflectance of floating algae. In addition, a color-based algorithm based on the relationship between CIE color space and FAI was developed to identify greenish algal blooms [12]. Machine learning algorithms, which consider both the spectral and spatial features of GOCI R_{rc} data, have been used to monitor floating macroalgae blooms [23], [24]. For floating bloom detection,

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the spectral wavelengths of each sensor determine the selection of the bloom detection index; for example, FAI can be used in Moderate Resolution Imaging Spectroradiometer (MODIS), The Visible and Infrared Imager/Radiometer Suite (VIIRS), and Operational Land Imager (OLI) data with the short-wave infrared (SWIR) band at approximately 1240 or 1640 nm, and AFAI was designed for satellite sensors, such as Geostationary Ocean Color Imager (GOCI) and Ocean and Land Color Instrument (OLCI), with wavelengths at 745 and 865 nm.

Multisensor observations, such as Thematic Mapper (TM), Enhanced Thematic Mapper Plus (ETM+), and OLI mounted on the Landsat satellites, of floating algae can provide information on the spatiotemporal dynamics of algal blooms over decades. As one of the longest satellites in orbit, AQUA MODIS (2002–present) has been the main data source for monitoring floating blooms in large lakes; however, AQUA MODIS (design life of 6 years) has shown associated degradation over these years [25]. It is important to continually ensure the accuracy of derived products and assess cross-sensor agreement [26]. Therefore, multisensor satellite data have been used in the assessment of bio-optical parameters in coastal and inland water, including the intercomparison of multisensors in TOA reflectance, remote sensing reflectance, water clarity [26], chlorophyll-a concentration (Chla) [27], Suspended Particulate Matter (SPM) [28], and algal bloom [29] using MODIS and VIIRS, as well as comparisons of Sentinel 2-MSI (MultiSpectral Instrument) and Landsat 8 OLI in deriving Total Suspended Matter (TSM) [30], [31], water clarity [32], turbidity [33], and algal bloom [34] in inland waters. Sentinel 2 MSI and Sentinel 3 OLCI have also been combined to estimate Chla in global lakes and coastal waters [35], [36].

In addition, the one- or two-day revisit times of MODIS, VIIRS, and OLCI have spatial resolutions that exceed 250 m; as a result, the percentage of blooms in the pixels with low spatial resolution can be estimated with pixel un-mixing methods [6], [7], [37]. The linear un-mixing methods in detecting floating blooms assume that the mixed pixel can be described as a linear combination of pure water and pure bloom. Several linear designed indexes, such as difference vegetation index (DVI), FAI, AFAI, and VB-FAH, were proven to be related well with FAC, pigment concentration, or biomass [37], [38], [39]. The determination of the lower bound and upper bound of bloom-containing pixels varied among different methods, including fixed water and pixel endmembers extracted from concurrent data with higher spatial resolution [6], searching the nearest pure water and bloom pixels within the image [40]. However, the linear decomposition methods did not consider the multiple scattering within the field of view, and they could not reflect the actual conditions of floating scums and remote sensing signals, which are asymptotically saturated with the accumulation of floating scums. Therefore, a more reasonable relationship between the floating algae cover (FAC) of a mixed pixel and the floating detection index considering the multiple scattering should be developed.

Based on the consistency of R_{rc} and the floating algae detection index of MODIS, VIIRS, GOCI, and OLCI, this study aims to 1) develop a FAC algorithm following a logistic (S-curve) function using OLI as a reference; 2) evaluate the performance of the FAC model in detecting floating algae blooms; 3) build a multisensor dataset of floating algae blooms in Lake Chaohu and Lake Taihu from 2002 to 2020; and 4) analyze and discuss the spatial-temporal dynamics of FAC.

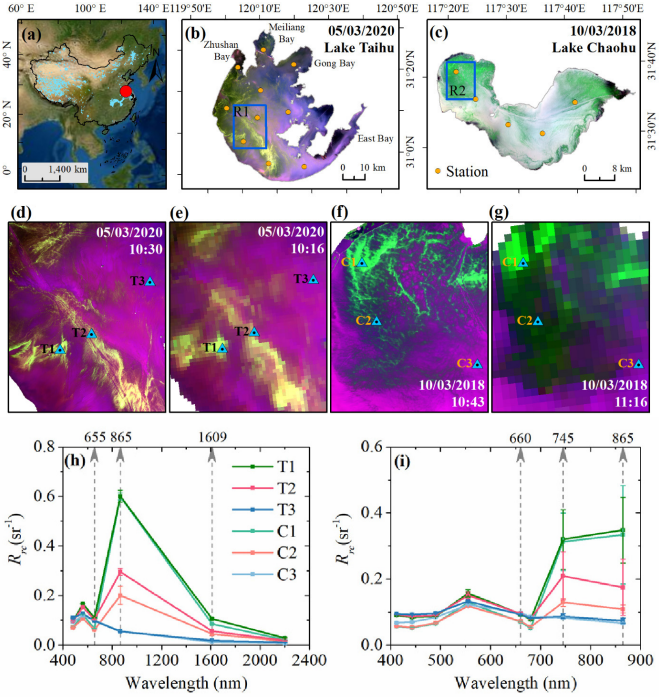


Fig. 1. (a) Location of Lake Taihu and Lake Chaohu. RGB images of (b) Lake Taihu on May 3, 2020, and (c) Lake Chaohu on October 3, 2018. Comparison of false-color RGB (FRGB, R: red, G: NIR, B: blue) from OLI and GOCI of zones (d) and (e) R1 and (f) and (g) R2. The R_{rc} spectra of stations (T1–T3, C1–C3) of OLI and GOCI are plotted in (h) and (i), respectively.

II. Data and Methods

A. Study Area

Lake Taihu (2338 km^2; mean depth, 1.9 m) and Lake Chaohu (768 km^2; mean depth, 2.7 m) are two shallow freshwater lakes located in the lower Yangtze River basin in China (see Fig. 1). These two lakes have suffered from severe eutrophication and frequent cyanobacterial blooms in recent decades, mainly due to climate variability, nutrients load, local rapid population growth, and economic development [41], [42]. In Lake Taihu, most of the nutrient-rich wastewater comes from the northern and northwestern regions. Harmful cyanobacterial scums regularly occur in the northern Meiliang, Zhushan, and Gonghu bays and in the western part of Lake Taihu [43]. Aquatic macrophytes, including emergent, floating, and submerged macrophytes, often appear in the eastern and southern parts of Lake Taihu [44]. The western part of Lake Chaohu has elevated eutrophication, and its mean concentrations of TP and TN were significantly higher than those in the eastern lake [45]. Floating algal blooms and phytoplankton pigments in the western part of Lake Chaohu were significantly higher than those in eastern Lake Chaohu [7], [45].

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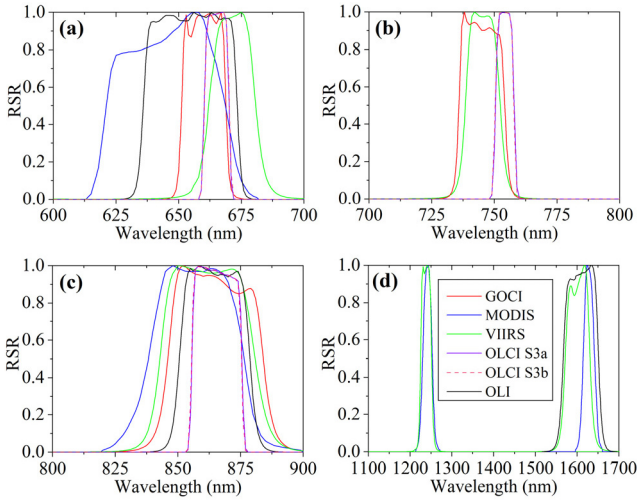


Fig. 2. RSR functions of MODIS, VIIRS, GOCI, OLCI, and OLI in (a) red, (b) and (c) NIR, and (d) SWIR spectral ranges.

B. Satellite Data Processing

MODIS aboard the AQUA satellite was launched in 2002, acquiring data in 36 spectral bands. Seven of these spectral bands, falling within the range from 469 to 2130 nm (250–500 m), were designed for land and atmosphere monitoring with a higher dynamic range. VIIRS, mounted on the Suomi National Polar-orbiting Partnership (S-NPP, launched in October 2011) spacecraft, is the successor to MODIS. VIIRS has 22 spectral bands ranging from 412 nm to 12 μm (375–750 m). GOCI resides aboard the Communication, Ocean and Meteorological Satellite (COMS) launched by the Korean Ocean Satellite Center (KOSC). It acquired data in eight spectral bands (six visible, two NIR) with a spatial resolution of approximately 500 m over the Korean Sea from June 2010 to March 2021. The OLCI, aboard Sentinel 3A (2016-05-15 to PRESENT) and Sentinel 3B (2018-04-25 to PRESENT), has a native spatial resolution of approximately 300 m with 21 spectral bands ranging from 400 to 1020 nm. Landsat-8 OLI (2013–present) with 30 m spatial resolution was used to derive the bloom percent of mixed pixels and validate the fractional FAC model. The center, width, and relative spectral response (RSR) functions of the spectral bands of each sensor used in this study are illustrated in Table I and Fig. 2.

Level-1 MODIS, VIIRS, and GOCI data were downloaded from NASA GSFC (http://oceancolor.gsfc.nasa.gov), OLCI data were downloaded from the European Space Agency (ESA) (https://scihub.copernicus.eu), and OLI data were downloaded from the United States Geological Survey (USGS) (https://earthexplorer.usgs.gov). Due to the failure of accurate atmospheric correction methods when applied to turbid waters, floating blooms, and aquatic macrophytes, R_{rc} was employed for processing in this study. MODIS, VIIRS, and GOCI Level-1 data were processed to Level-2 R_{rc} data using SeaDAS (version 7.5.3). OLCI and Landsat 8 OLI data were processed using the latest ACOLITE software package (version 20220222.0, https://github.com/acolite) [46]. In order to compare the multisensor data, the Level-2 R_{rc} data of

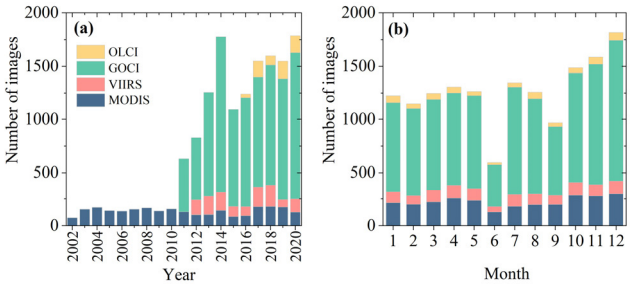


Fig. 3. Number of images of MODIS, VIIRS, GOCI, and OLCI cover Lake Taihu and Lake Chaohu in different (a) years and (b) months.

MODIS, VIIRS, GOCI, and OLCI were resampled to 250 m, and OLI R_{rc} data were resampled to 25 m. The $R_{rc}(\lambda)$ data were used to compose Red-Green-Blue (RGB) images and false-color RGB (FRGB, R: red, G: NIR, B: blue) images. For the period of 2002–2020, 2676, 1255, 10 679, and 637 relatively cloud-free MODIS, VIIRS, GOCI, and OLCI scenes were used for subsequent analysis of Lake Taihu and Lake Chaohu. The annual and monthly number of available images of each sensor are presented in Fig. 3. Using red and SWIR bands to form a baseline that was then subtracted from the NIR band, an FAI was designed for MODIS [16]

$$\mathrm{FAI} = R_{rc}(\lambda_{\mathrm{NIR}}) - R_{rc}(\lambda_{\mathrm{RED}}) - (R_{rc}(\lambda_{\mathrm{SWIR}}) - R_{rc}(\lambda_{\mathrm{RED}})) * \frac{(\lambda_{\mathrm{NIR}} - \lambda_{\mathrm{RED}})}{(\lambda_{\mathrm{SWIR}} - \lambda_{\mathrm{RED}})} \quad (1)$$

where $\lambda_{\mathrm{RED}} = 645$, $\lambda_{\mathrm{NIR}} = 859$, and $\lambda_{\mathrm{SWIR}} = 1240$ nm. FAI can also be used for VIIRS data with λ_{RED}, λ_{NIR}, and λ_{SWIR} at 671, 862, and 1238 nm, respectively. For OLI, $\lambda_{\mathrm{RED}} = 655$, $\lambda_{\mathrm{NIR}} = 865$, and $\lambda_{\mathrm{SWIR}} = 1609$ nm.

The baseline subtraction approach makes this approach robust when applied to waters suffering from aerosols, thin clouds, and sun glint in detecting floating algal blooms [16]; however, FAI cannot be applied to GOCI and OLCI directly due to the lack of the SWIR band. A NIR band was used to replace the SWIR band, and the AFAI was built [6]

$$\mathrm{AFAI} = R_{rc}(\lambda_{\mathrm{NIR1}}) - R_{rc}(\lambda_{\mathrm{RED}}) - (R_{rc}(\lambda_{\mathrm{NIR2}}) - R_{rc}(\lambda_{\mathrm{RED}})) * \frac{(\lambda_{\mathrm{NIR1}} - \lambda_{\mathrm{RED}})}{(\lambda_{\mathrm{NIR2}} - \lambda_{\mathrm{RED}})}. \quad (2)$$

Here, the λ_{RED}, λ_{NIR1}, and λ_{NIR2} of GOCI are at 660, 745, and 865 nm, respectively. For OLCI, λ_{RED}, λ_{NIR1}, and λ_{NIR2} are at 665, 754, and 865 nm, respectively.

To eliminate the influence of aquatic macrophytes, cyanobacteria and macrophyte index (CMI) and turbid water index (TWI) were combined to extract macrophytes and turbid waters [44] in MODIS and VIIRS data. The SWIR band of CMI was changed to 865 and 1016 nm in GOCI and OLCI data, respectively (see details in Supporting Information). The CMI, and TWI threshold value of each sensor were illustrated in Table S.1. Emergent/floating macrophytes and floating blooms have similar spectral characteristics, making them difficult to distinguish in satellite images. Considering the natural phonological characteristics of aquatic vegetation and algal blooms, the time series of pixels from May to October and frequencies larger than 75% were regarded as emergent/floating macrophytes [47].

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TABLE I
CENTER AND WIDTH OF THE SPECTRAL BANDS OF THE SATELLITE SENSORS IN RED, NIR, AND SWIR BANDS

	MODIS (AQUA)		VIIRS (SNPP)		GOCI (COMS)		OLCI (Sentinel 3)		OLI (Landsat 8)	
band	Center (nm)	Width (nm)	Center (nm)	Width (nm)	Center (nm)	Width (nm)	Center (nm)	Width (nm)	Center (nm)	Width (nm)
red	645	47.5	671	18.8	660	17.7	665	10.0	655	37.5
NIR1	/	/	745	13.9	745	18.3	754	7.5	/	/
NIR2	859	38.3	862	38.1	865	38.1	866(3A) 865(3B)	20.0	865	28.2
SWIR1	1240	23.4	1238	26.4	/	/			/	/
SWIR2	/	/	/	/	/	/			1609	84.7

C. Match-Up Pairs of Multisensor Data

Matched pairs between any of the two satellite data within 1 h were selected as the concurrent images. Ten virtual stations in Lake Taihu and five virtual stations in Lake Chaohu [see Fig. 1(b) and (c)] were chosen for comparison of the R_{rc} values of the red, NIR, SWIR bands, and FAI or AFAI values of concurrent images. The comparison was performed based on common pixels in which both of the compared sensors showed valid observations. We assumed that within 1 h, algae features did not change significantly, thus allowing for a direct comparison. Restrictions were used to distinguish the matched pairs: 1) coefficient of variation (CV) <10%; 2) sun zenith angle (SZA) <70°; and 3) the difference in overpass time (Δt) within 1 h.

D. Model Development

The overpass time of OLI in Lake Taihu and Lake Chaohu is approximately 10–11 A.M. local time; however, the MODIS and VIIRS data over Lake Taihu and Lake Chaohu used in this study were recorded from afternoon passes, which did not meet the matched criteria above. Therefore, concurrent Landsat 8 OLI and GOCI images (overpass time ±1 h, cloudless) were used to develop the FAC model. First, OLI spatial degradation considering the point spread function (PSF) was conducted according to the method in Che et al. [48]. Second, the FAC model was developed based on the bloom percent of degraded OLI and AFAI (see Fig. 4).

1) OLI Spatial Degradation Considering PSF: The spatial degradation of OLI was undertaken by modeling the PSF [48]

$$L_{250\,\mathrm{m}}(\lambda) = \mathrm{subsample}(\mathrm{psf}(\lambda) * L_{25\,\mathrm{m}}(\lambda)) \tag{3}$$

where $L_{250\,\mathrm{m}}(\lambda)$ is the degraded OLI image at 250 m for wavelength λ, $L_{25\,\mathrm{m}}(\lambda)$ is the preprocessed 25 m OLI R_{rc} reference image for wavelength λ, psf(λ) is the $n \times n$ convolution kernel matrix, $*$ denotes convolution. The spatial resolution of MODIS, VIIRS, GOCI, and OLCI was resampled to 250 m in this study, and the OLI data were resampled to 25 m, so the subsample is subsampled every 10 (=250/25 m) pixels from the convoluted image. n is set to 41 such that 41 × 41 resampled OLI pixels (1.025 × 1.025 km^2) are used in the convolution assuming that the adjacent effect is negligible beyond 20 pixels [49], [50]

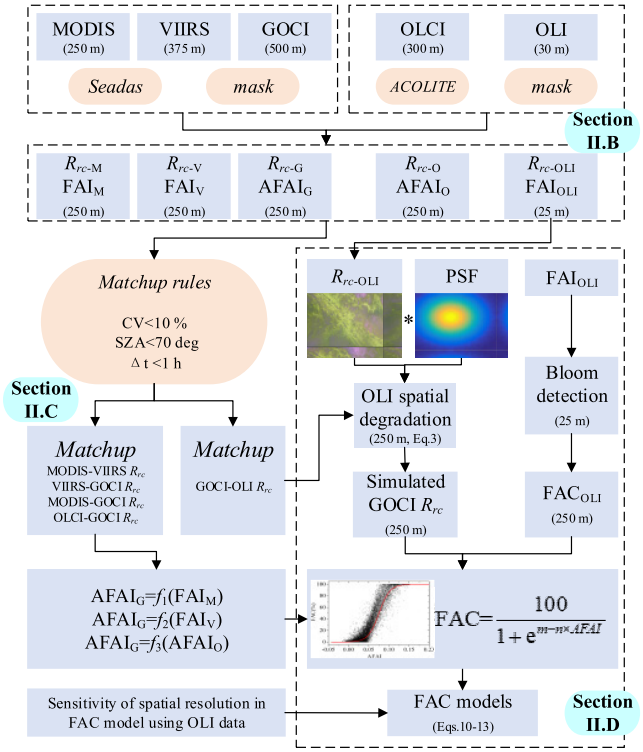


Fig. 4. Flowchart of the data processing and FAC model development.

$$\mathrm{psf}(i, j, \sigma_\lambda) = \frac{1}{2\pi\sigma_\lambda^2} \exp\left(-\frac{(i-21)^2 + (j-21)^2}{2\sigma_\lambda^2}\right) \tag{4}$$

$$\sigma_\lambda = \arg\max R^2(L_{250\,\mathrm{m}}(\lambda, \sigma_\lambda), S_{250\,\mathrm{m}}(\lambda, \sigma_\lambda)). \tag{5}$$

$S_{250\,\mathrm{m}}$ represents the R_{rc} of the GOCI image, σ_λ equals spatial resolution divided by 25 m. The PSF of each sensor was determined by maximizing the correlation coefficient (r) and minimizing the relative error (δ) between the degraded OLI and GOCI images, and then $L_{250\,\mathrm{m}}$ was calculated using (3).

There were 17 concurrent images between OLI and GOCI that were used to determine the σ_λ. OLI and GOCI had different spectral bands, so we compared their blue, green, red, and NIR bands respectively. Fig. 5 shows the variations of mean r and δ with increasing spatial resolution in blue, green, red, and NIR R_{rc} bands of GOCI (490, 555, 660, and 865 nm) and OLI (483, 561, 655, and 865 nm). Due to the lack of a SWIR band in GOCI data, the FAI, and AFAI could not compare directly and assumed a band independent value

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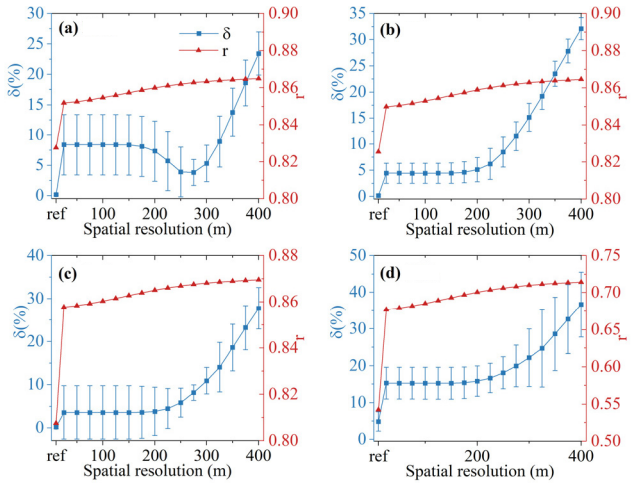


Fig. 5. δ and r between the degraded OLI R_{rc} (483, 561, 655, and 865 nm) and GOCI R_{rc} (490, 555, 660, and 865 nm) as a function of the GOCI PSF standard deviation σ_λ for (a) blue, (b) green, (c) red, and (d) NIR bands.

of σ_λ. The mean r and δ had better performance in spatial resolution around 200–250 m in blue, green, red, and NIR R_{rc} bands of GOCI and OLI (see Fig. 5), and δ increased obviously when the spatial resolution became larger than 200 m. Thus, σ_λ was assumed to be band independent and was set to be 8 (=200/25 m). The uncertainties introduced by spatial resolution and spectral differences among the sensor would be discussed further in Section IV-B. A simulated AFAI of GOCI was derived using the $L_{250\,m}$ and its relationship with $S_{250\,m}$ derived AFAI.

2) Fractional FAC Model: Floating bloom pixels of the 25 m OLI images were extracted using the maximum gradient method [43], and the 25-m bloom of OLI was resampled to 250 m to calculate the bloom percent. The corresponding bloom percent (=number of bloom pixels/total pixels × 100%) at 250 m was regarded as the true value of FAC in each pixel.

The FAC of each pixel of the GOCI is calculated using the derived OLI bloom percent as a reference. The relationship between the simulated AFAI and FAC was determined using their matched pairs. The FAC functions can be modeled as a logistic regression model as follows:

$$\mathrm{FAC} = \frac{100}{1 + e^{m - n \times \mathrm{AFAI}}} \tag{6}$$

where 100 represents the maximum value of bloom cover (100%) in a pixel, and m and n were derived using curve fitting tools in MATLAB 2018R. For MODIS and VIIRS, the relationship between its FAI and AFAI of GOCI was first determined, and then the FAC value was calculated following (6), respectively. The integrated equivalent bloom area (EBA, km^2) of the lake is calculated as

$$\mathrm{EBA} = \sum_{i=1}^{NP} \mathrm{FAC}(i) \times 0.25 \times 0.25 \tag{7}$$

where NP is the total number of pixels, and 0.25 km is the pixel spatial resolution.

The linear interpolation model was provided to calculate the bloom coverage within the pixels of GOCI data, as follows:

$$\alpha_i = \frac{\mathrm{AFAI\text{-}AFAI_L}}{\mathrm{AFAI_H} - \mathrm{AFAI_L}} \times 100\% \tag{8}$$

where α_i is between 0 and 100%, and the lower bound ($\mathrm{AFAI_L}$) and upper bound ($\mathrm{AFAI_H}$) of bloom-containing pixels were −0.001 and 0.12 in Lake Taihu [6] (Linear-1 model). Following the study of Cui et al. [37] (Linear-2 model), the upper bound was the maximum value of AFAI over the algae pixels, and the lower bound was the maximum value over the water region in each image. Another spectral un-mixing method [40] (Linear-3 model) that integrates the spatial and spectral characteristics of image data during the process of endmember extraction was also used in this study to evaluate the model performance.

E. Performance Metrics

Several statistical metrics were used to compare the differences between satellite-derived products, such as R_{rc} data, FAI, FAC, and bloom area. These metrics included the coefficient of determination (R^2), mean relative error (δ), unbiased percentage difference (UPD), root mean square error (RMSE), and CV

$$\delta = \frac{1}{N} \sum_{i=1}^{N} \frac{|X_i - Y_i|}{X_i} \times 100\% \tag{9}$$

$$\mathrm{UPD} = \frac{1}{N} \sum_{i=1}^{N} \frac{|X_i - Y_i|}{X_i + Y_i} \times 200\% \tag{10}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (X_i - Y_i)^2} \tag{11}$$

$$\mathrm{CV} = \frac{\sqrt{\frac{1}{\mathrm{NS}} \frac{1}{\mathrm{NS}} \sum_{j=1}^{\mathrm{NS}} \sum_{i=1}^{\mathrm{NS}} \left(P_{i,j} - P_m\right)^2}}{P_m} \tag{12}$$

where N is the number of data pairs, X_i and Y_i are the ith data pairs, $P_{i,j}$ is the value of the ith and jth elements in an $S \times S$ matrix, P_m is the mean value of the product for the $S \times S$ matrix, and NS is the number of elements in the $S \times S$ matrix, S is set to be 3 in this study.

III. Results

A. Consistency of R_{rc}, FAI, AFAI of Matched Pairs

Figs. 6 and 7 show the comparison of the R_{rc} values and FAIs (FAI, AFAI) of the multisensor data at the 15 virtual stations in Lake Taihu and Lake Chaohu. The statistical summary before and after using data quality control criteria is provided in Table II. According to the restrictions rules in Section II-D, a total of 2196 matched pairs of MODIS and VIIRS data were selected to compare the R_{rc} values in the red, NIR, SWIR bands, and FAI values [see Figs. 6(a)–(c) and 7(a)]. Overall, the matched pairs showed consistency among the three R_{rc} bands with no obvious overestimation or underestimation. The R_{rc}(RED) of MODIS is centered at 645 nm [$R_{rc-M}(645)$] with a width of 47.5 nm, and the R_{rc}(RED) of VIIRS is centered at 671 nm

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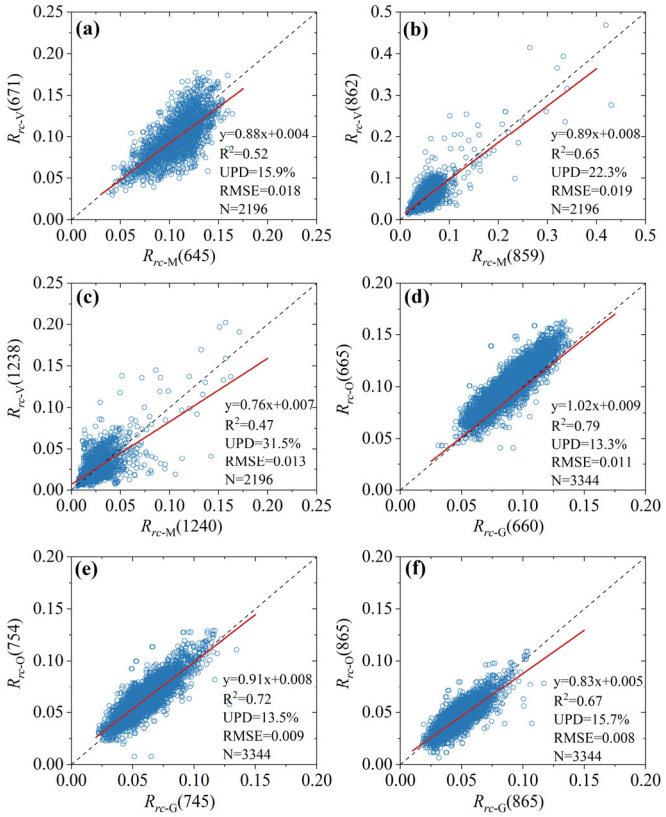


Fig. 6. Comparisons of R_{rc} of (a)–(c) MODIS and VIIRS, (d)–(f) OLCI and GOCI in the red, NIR, and SWIR bands. The matched pairs were extracted based on the match-up rules from ten virtual stations in Lake Taihu and five virtual stations in Lake Chaohu.

$[R_{rc-V}(671)]$ with a width of 18.8 nm. The difference in center wavelength and bandwidths contributed to the variation between $R_{rc-M}(645)$ and $R_{rc-V}(671)$ ($R^2 = 0.52$, slope $= 0.88$, intercept $= 0.004$). Even though SWIR bands had the lowest consistency, the FAI of MODIS and FAI of VIIRS had a higher consistency with $R^2 = 0.78$ and UPD $= 21.9\%$. A previous study showed that MODIS and VIIRS had a higher agreement when the SWIR of R_{rc} was subtracted from NIR in Lake Hongze in China [28].

A total of 3344 matched pairs between GOCI and OLCI were compared in the red and two NIR bands, as well as AFAI [see Figs. 6(d)–(f) and 7(b)], and the pairs had better agreement than those of MODIS and VIIRS. The main reason was that the GOCI and OLCI data had similar band centers and bandwidths, which could avoid spectral distortion. We also compared the matched pairs between the AFAI of GOCI ($AFAI_G$) and FAI of MODIS (FAI_M) and between the AFAI of GOCI and FAI of VIIRS (FAI_V). They had R^2 values greater than 0.80 and lower RMSEs [see Fig. 7(c) and (d)]. The low slope of approximately 0.6 and relatively high UPD (>54.9%) between FAI and AFAI were associated with their definitions, as the lower baseline of FAI produced a higher difference value than that of AFAI.

B. Performance of the FAC Model of Multisensor Data

The proposed FAC model of GOCI follows a logistic curve with a maximum value of 100% and a minimum

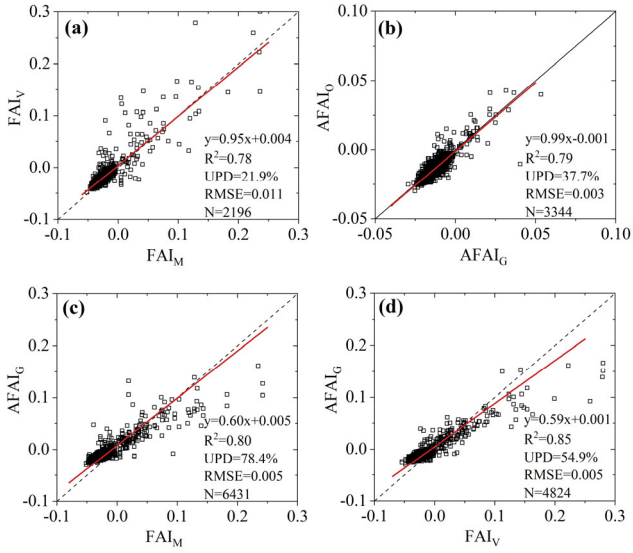


Fig. 7. Comparisons of (a) FAI of MODIS and FAI of VIIRS, (b) AFAI of GOCI and AFAI of OLCI, (c) FAI of MODIS and AFAI of GOCI, (d) FAI of VIIRS and AFAI of GOCI. The matched pairs were extracted based on the match-up rules from ten virtual stations in Lake Taihu and five virtual stations in Lake Chaohu.

value of 0% ($R^2 = 0.93$, RMSE $= 4.13$, $N = 85634$) [see Figs. 8(a) and (13)]. From the comparison of the linear and FAC models, we can see that the linear model had obvious overestimation in the ranges of low AFAI and underestimation in the high AFAI values. Statistically, in waters with a majority part of non-bloom pixels, the linear model would have an overall overestimation of FAC and EBA. In order to decrease the influence of data distribution on function fitting, we randomly selected samples with nearly even distribution, and the result indicated that the logistic curve still had better performance than linear fitting [see Fig. 8(b)].

The FAC models of MODIS, VIIRS, and OLCI are derived from the relationships between $AFAI_G$ and FAI_M, FAI_V, and $AFAI_O$, respectively. The FAC models for GOCI, MODIS, VIIRS, and OLCI are as follows:

$$FAC_G = \frac{100\%}{1 + e^{5.0-73.6AFAI_G}} \tag{13}$$

$$FAC_M = \frac{100\%}{1 + e^{4.6-44.2FAI_M}} \tag{14}$$

$$FAC_V = \frac{100\%}{1 + e^{4.9-43.4FAI_V}} \tag{15}$$

$$FAC_O = \frac{100\%}{1 + e^{4.9-74.3AFAI_O}}. \tag{16}$$

Fig. 9 illustrates the bloom density of the FAC model and three linear models (Linear-1, Linear-2, and Linear-3) of Lake Taihu on May 3, 2020. The bloom area extracted from OLI was 185.9 km^2 at 10:30 A.M. [see Fig. 9(a)], and the EBA values of the four models derived from GOCI data were 206.4, 334.9, 143.1, and 308.5 km^2 at 10:16 AM see Fig. 9(c)]. The bloom densities derived from Linear-1 and Linear-3 were obviously greater than that of the FAC model, especially in regions with low bloom density. The EBA derived by the Linear-1 model was approximately 1.75× the EBA derived from the FAC model. Comparatively, the Linear-2 model had a relatively

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TABLE II

SUMMARY OF THE RESULTS FOR THE MATCHED PAIRS FOR MODIS, VIIRS, GOCI, AND OLCI; FOR MODIS AND VIIRS, FAI WAS CALCULATED USING (1); FOR GOCI AND OLCI, AFAI WAS CALCULATED USING (2)

Sensors	Standard Deviation		UPD (%)	Samples	Restrictions
	MODIS FAI	VIIRS FAI			
MODIS VS. VIIRS	0.029	0.030	55.6	4955	No restrictions
	0.021	0.024	21.9	2196	CV<10%, SZA<70 deg, Δh <1h
	OLCI AFAI	GOCI AFAI			
OLCI VS. GOCI	0.021	0.026	49.5	7480	No restrictions
	0.005	0.006	37.7	3344	CV<10%, SZA<70 deg, Δt <1h
	MODIS FAI	GOCI AFAI			
MODIS VS. GOCI	0.026	0.017	82.7	11545	No restrictions
	0.016	0.011	78.4	6431	CV<10%, SZA<70 deg, Δt <1h
	VIIRS FAI	GOCI AFAI			
VIIRS VS. GOCI	0.017	0.029	260.2	10065	No restrictions
	0.013	0.020	54.9	4824	CV<10%, SZA<70 deg, Δt <1h

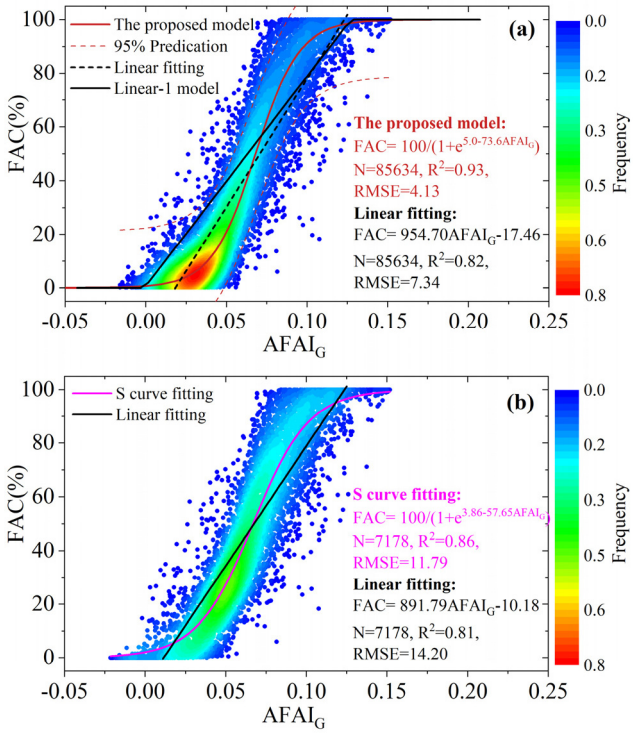


Fig. 8. (a) FAC model developed using matched images between OLI bloom percent and simulated GOCI AFAI at 250 m. The linear pixel un-mixing model (Linear-1) of bloom density was derived using (8). (b) *S* curve fitting and linear fitting using samples with nearly even distribution.

higher mean lower bound (0.005 for GOCI) and upper bound (0.22 for GOCI) of bloom-containing pixels, making the low values of FAC on May 3, 2020, in Lake Taihu.

The UPD of FAC in each matched pair was mapped to show the distribution of variability of FAC (see Fig. 10). Higher variability was observed in the MODIS-GOCI and VIIRS-GOCI matched pairs in the western part of Lake Taihu. The higher difference along the lake boundary was mainly caused by the land adjacency effect and the variability in spatial resolution of different satellite images.

C. Validation of the FAC Model Using Matched Data

The EBA of the matched pairs of OLI-GOCI and OLI-OLCI were compared, and the results indicated that the R^2 was 0.96 and 0.76, and the UPD was 37.6% and 39.2%, respectively [see Fig. 11(a)]. The large deviation in the range with low OLI bloom area demonstrated that satellite data with low spatial resolution may not be able to detect the small patches of floating blooms. Due to the time gap (larger than 3 h), the relationships of EBA in OLI-MODIS and OLI-VIIRS are nonsignificant with $R^2 < 0.44$ and p-value >0.12 (data not shown). MODIS and VIIRS had good agreement ($R^2 > 0.74$, UPD $< 42.2\%$) with matched GOCI data, which also indicated their agreement with OLI indirectly [see Fig. 11(b)].

Considering the influence of the number of valid pixels on EBA, concurrent images with similar total valid pixels (larger than 80%) were chosen to compare their EBA. The matched pairs between different sensors showed that high consistency of EBA, especially in ranges with EBA > 10 km^2, was acquired with $R^2 > 0.87$ and UPD $< 46.46\%$ (see Fig. 12). In cases with EBA larger than 10 km^2, the UPD was decreased to 29.6%, 31.87%, 28.53%, and 25.02% in Fig. 12(a)–(d), respectively. Note that the EBA of VIIRS had a negative intercept with that of GOCI and MODIS, which may be caused by the coarse spatial resolution and lower SNRs of VIIRS [51].

D. Spatial-Temporal Dynamics of FAC

The EBA of multiple sensors, monthly maximum EBA of multiple sensors, monthly maximum EBA of MODIS, and daily bloom area of OLI in Lake Taihu and Lake Chaohu are illustrated in Fig. 13(a) and (b). The combined multisensor FAC data were then used to analyze the spatial-temporal dynamics of EBA in Lake Taihu and Lake Chaohu from 2002 to 2020. The annual EBA showed a nonmonotonic and overall increasing trend in Lake Taihu and Lake Chaohu

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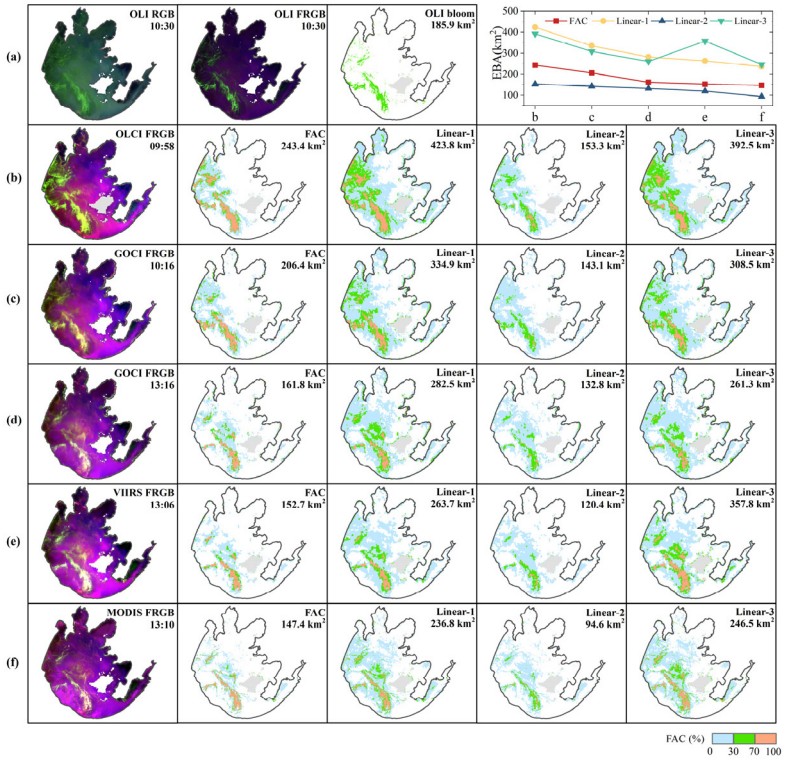


Fig. 9. Comparisons of the bloom density of Lake Taihu derived from the FAC model and the three linear models (Linear-1, Linear-2, Linear-3) using different sensors on May 3, 2020 (a) OLI at 10:30, (b) OLCI at 09:58, (c) GOCI at 10:16, (d) GOCI at 13:16, (e) VIIRS at 13:06, (f) MODIS at 13:10. The time is UTM+8h. The OLI bloom was extracted using the maximum gradient method. EBA of images in (b–f) derived using different models were also plotted.

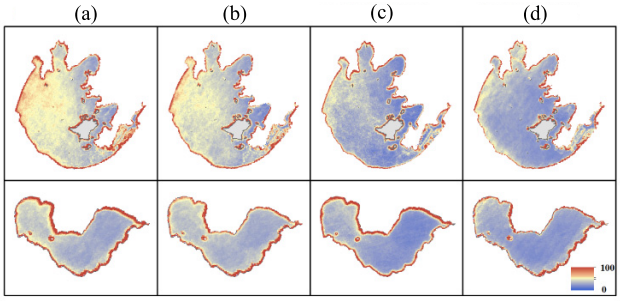


Fig. 10. Distribution of the mean UPD of FAC in Lake Taihu and Lake Chaohu derived from matched pairs: (a) MODIS versus GOCI, (b) VIIRS versus GOCI, (c) MODIS versus VIIRS, and (d) OLCI versus GOCI.

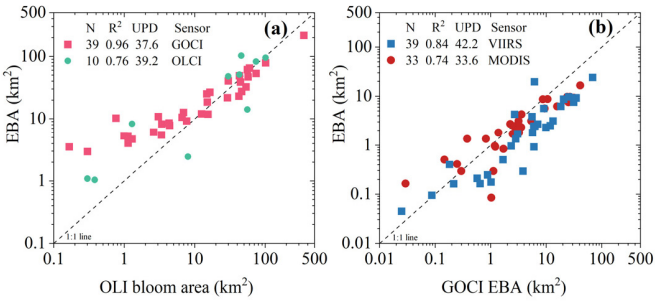


Fig. 11. (a) Validation of bloom area derived from GOCI and OLCI data using the concurrent OLI data and (b) comparison of bloom area of VIIRS and MODIS from GOCI matched pairs within 1 h. Note that the delta time between OLI and MODIS and VIIRS is usually larger than 3 h, and we used the concurrent GOCI data as the reference in (b).

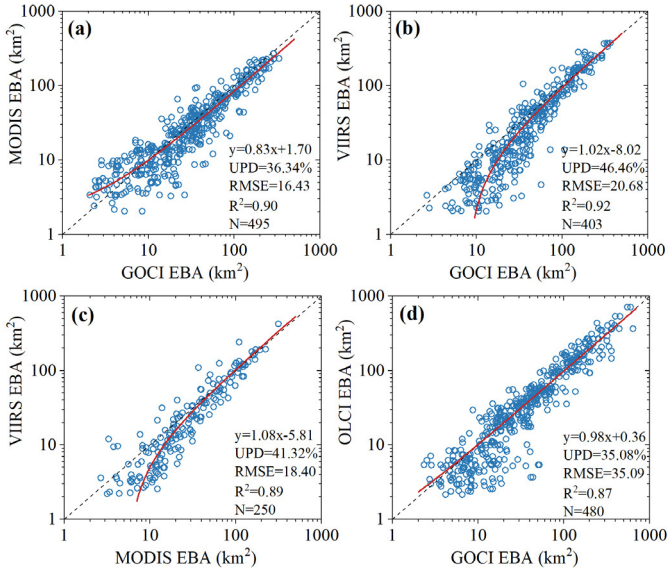


Fig. 12. Comparisons of concurrent EBA of (a) GOCI and MODIS, (b) GOCI and VIIRS, (c) MODIS and VIIRS, and (d) GOCI and OLCI.

[see Fig. 13(c) and (d)]. There was an obvious increase between the 2002–2010 and 2011–2020 periods, with the slope of the annual averaged EBA being 2.87 and 0.67 in Lake Taihu and Lake Chaohu, respectively. The EBA of Lake Taihu reached a maximum value in 2017, which was 3.5× the mean EBA in 2007. Field measurements also showed the maximum value of Chla in 2017 (2007–2017) [42]. Compared with using MODIS data only, the combined multisensor data monitored overall EBA values that were 45% and 23% higher from 2002 to 2020 in Lake Taihu and Lake

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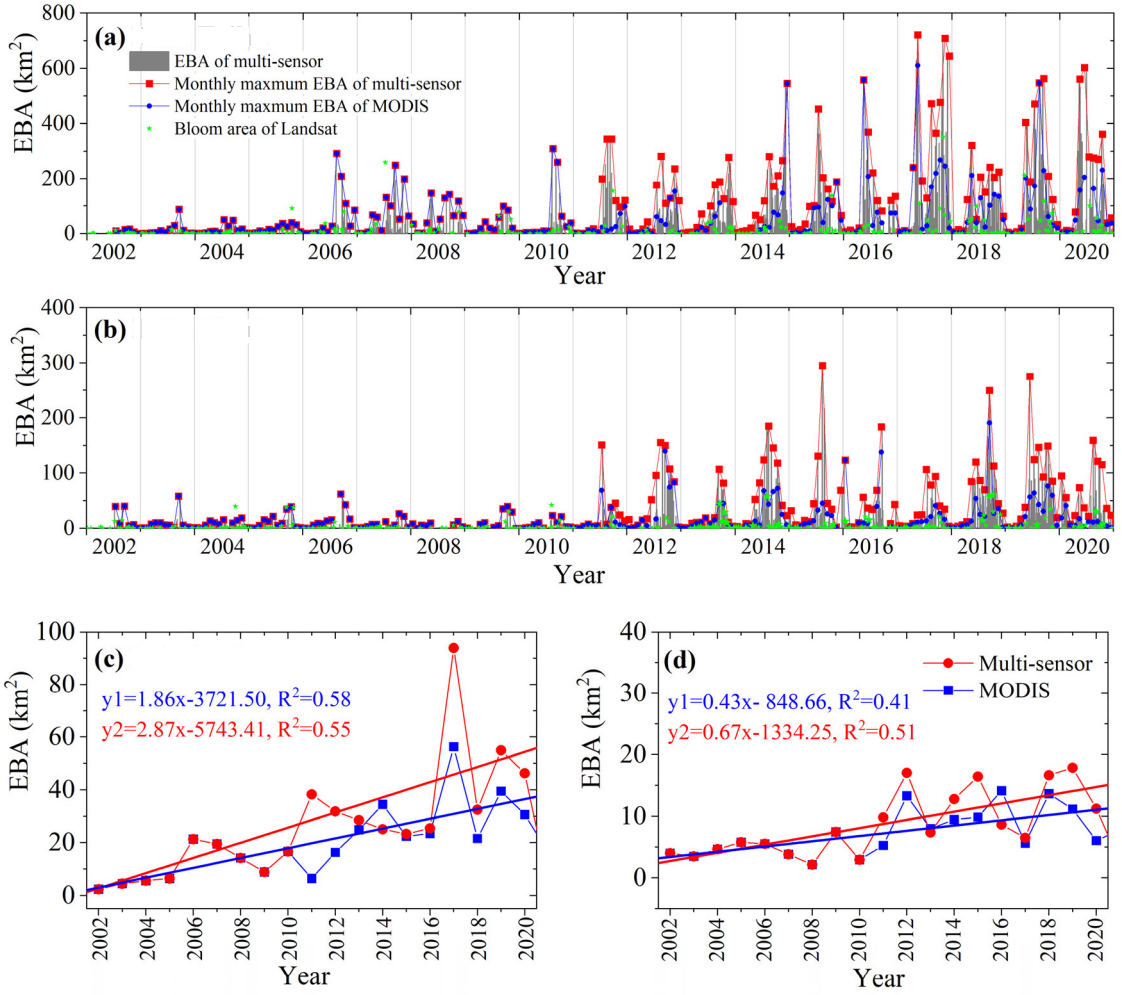


Fig. 13. Time series of the EBA of multisensor data and monthly maximum EBA of multisensor and MODIS data, bloom area of Landsat data in (a) Lake Taihu and (b) Lake Chaohu. The yearly mean EBA of multisensor and MODIS data in (c) Lake Taihu and (d) Lake Chaohu.

Chaohu, respectively. The EBA of Lake Chaohu first showed a slight increase in 2005, followed by an obvious increase in 2012 (from 2002 to 2020), which is in accordance with a previous study [7]; furthermore, the EBA was relatively high from 2018 to 2020.

Figs. 14 and 15 illustrate the yearly and monthly mean FAC derived from the multisensor data. Before 2005, high FAC values were observed in Meiliang Bay and Zhushan Bay of Lake Taihu and the western part of Lake Chaohu. After 2010, the bloom often occupied the north and northwestern regions in Lake Taihu, which is in accordance with the distribution of cyanobacterial blooms [42], Chla [52], and PC (phycocyanin pigment) [53]. The middle part of Lake Chaohu exhibited a higher bloom frequency during summer and autumn after 2018, suggesting elevated eutrophication of the whole lake.

Diurnal changes in normalized EBA (=EBA of each hour/maximum EBA of the day) in Lake Taihu and Lake Chaohu are observed from the multisensor image series from 2011 to 2020 (see Figs. 16 and 17). Table III shows the statistics of the three types of diurnal distribution of normalized EBA in Lake Taihu and Lake Chaohu from 2011 to 2020. The days that had more than six measurements were chosen for curve fitting, among which 303 and 233 days were found to be significant in Lake Taihu and Lake Chaohu, respectively. Similar to a previous study in Lake Taihu [6], three typical patterns are illustrated in Fig. 17. Type 1 increased over the day from 8:00 to 15:00, Type 2 had a peak value at approximately 11:00 and then decreased, and Type 3 showed a near-continuous decrease. There is a clear difference between the seasonality of the three types, indicating the possible driving force of temperature in diurnal changes of bloom. For example, Type 1 occurred around Day 259 (±130) and Day 265 (±126) in Lake Taihu and Lake Chaohu, Type 3 occurred around Day 217 (±77) and Day 214 (±80) in Lake Taihu and Lake Chaohu.

IV. Discussion

A. Advantages of the Proposed FAC Model

Through the evaluation of the consistency of GOCI, MODIS, VIIRS, and OLCI in floating bloom detection, a nonlinear FAC model was developed, and then applied to the multisensor data. In the first step of the FAC model, PSF was modeled following a Gaussian function with parameters

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TABLE III
STATISTICS OF THE THREE TYPES OF DIURNAL CHANGE PATTERNS OF THE NORMALIZED EBA OF LAKE TAIHU AND LAKE CHAOHU

Lake name	Type	N	Percent (%)	Time with maximum EBA(h)	Julia day (d)	Maximum EBA (km^2)
Taihu	Type 1	59	19.5	14.0 ± 0.8	259 ± 130	58.94 ± 77.90
	Type 2	155	51.1	11.0 ± 0.7	249 ± 86	95.77 ± 111.53
	Type 3	89	29.4	8.5 ± 0.5	217 ± 77	113.46 ± 126.94
Chaohu	Type 1	52	22.3	14.0 ± 0.9	265 ± 126	22.08 ± 23.90
	Type 2	136	58.4	10.7 ± 0.7	227 ± 98	29.17 ± 36.22
	Type 3	45	19.3	8.6 ± 0.5	214 ± 80	48.41 ± 60.28

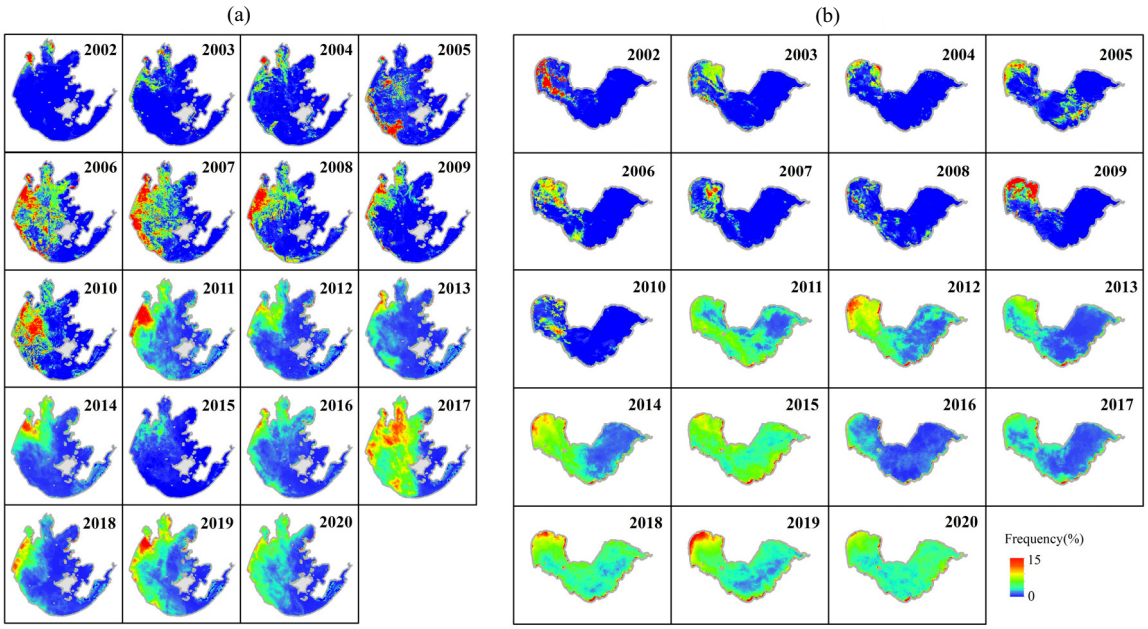


Fig. 14. Yearly mean FAC of multisensor data from 2002 to 2020 in (a) Lake Taihu and (b) Lake Chaohu.

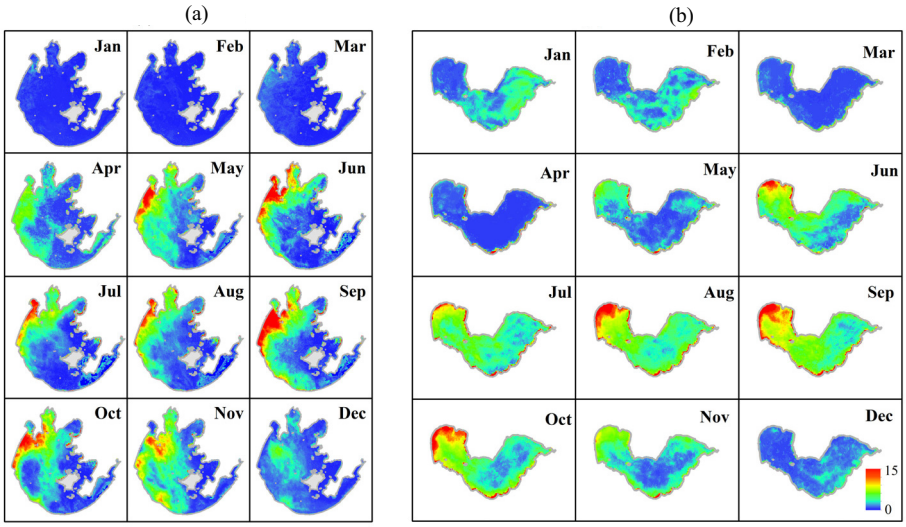


Fig. 15. Monthly mean FAC of multisensor data from 2002 to 2020 in (a) Lake Taihu and (b) Lake Chaohu.

estimated empirically by comparing the spatially degraded OLI R_{rc} and GOCI R_{rc} images. After pixel un-mixing, the EBA of multisensor calculated using the proposed FAC model was in accordance with the OLI bloom area.

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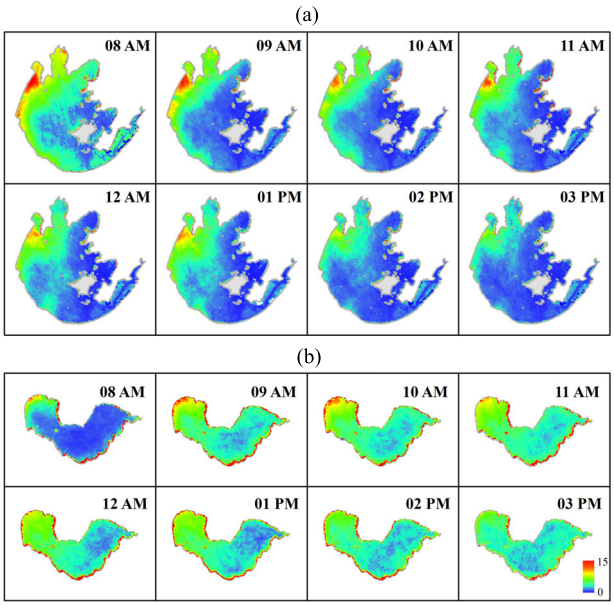


Fig. 16. Diurnal distribution of FAC derived from multisensor data (2011–2020) in (a) Lake Taihu and (b) Lake Chaohu.

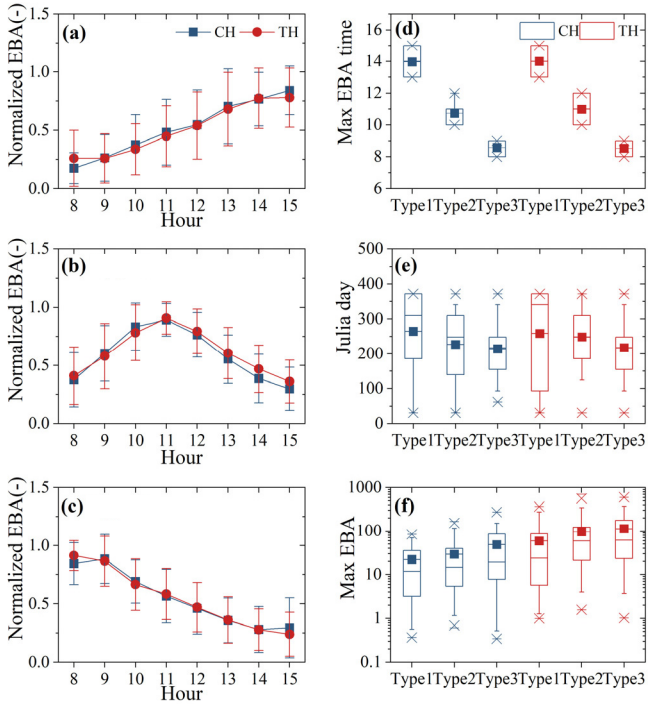


Fig. 17. (a)–(c) Diurnal changes of normalized EBA (=EBA/maximum EBA of the day) in Lake Taihu (TH) and Lake Chaohu (CH). Box plots of (d) time with maximum EBA, (e) Julia day, and (f) maximum EBA of the three types.

It demonstrated that the FAC model could improve the overestimation of bloom area without pixel un-mixing in low-spatial resolution images. EBA and BCA (bloom-containing area) in Lake Taihu and Lake Chaohu showed significant relationships with $R^2 > 0.83$ and $p < 0.01$ (see Fig. 18). In conditions with low bloom area, the EBA was approximately 4× as large as the BCA, whereas the EBA was approximately 2–2.5× as large as the BCA in conditions with

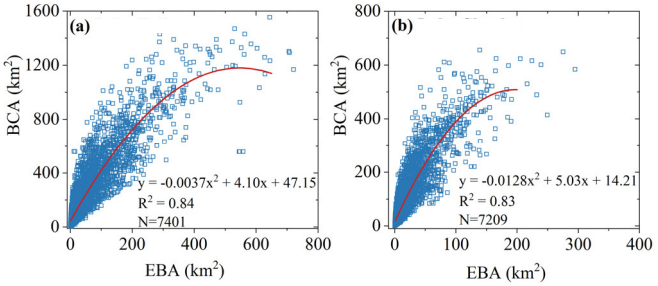


Fig. 18. Relationship between EBA and BCA in (a) Lake Taihu and (b) Lake Chaohu.

a large bloom area. However, it was different from the results derived using the FAI or AFAI threshold in MODIS or GOCI. For instance, the floating bloom area derived by Hu et al. [43] (BA_{Hu2010}) using $FAI_{MODIS} = -0.004$ from 2002 to 2008 was highly related to EBA and BCA in this study, with R^2 0.84 ($BA_{Hu2010} = 19.48EBA + 132.85$) and 0.93 ($BA_{Hu2010} = 8.72BCA + 76.33$), respectively. In addition, the previous study also indicated an approximate overestimation of 2.6× by MODIS (250-m resolution) relative to HJ-1 CCD (30-m resolution) in bloom detecting [37].

The linear-designed floating blooms indexes were often linearly related to bloom density or biomass in the previous studies [6], [37]. In fact, with the accumulation of floating blooms, the NIR spectral band becomes saturated, and the FAC reaches 100% asymptotically. Taking into account multiple scattering, we found that the FAC of low-spatial-resolution images increased with the AFAI following the logistic model (S-curve function). The proposed FAC model is theoretically closer to the actual conditions. In addition, this study also demonstrated that the bloom density or EBA is more sensitive to the determination of the lower and upper bound of AFAI or FAI in the linear pixel un-mixing model. The proposed FAC model increase slowly in the low and high FAC ranges, which could avoid the uncertainties introduced by the determination of the lower and upper bounds of AFAI in the linear pixel un-mixing model.

B. Uncertainties and Limitations in Model Development

The proposed FAC model was built based on the assumption that the low-middle spatial resolution is resampled to 250 m, and the FAI or AFAI of the multisensory data correlated well. In fact, the variations from band settings, spectral resolutions, and spatial resolutions may lead to uncertainties. The difference in band setting and spectral resolution of the multisensory data made it difficult to compare their corresponding bands directly. The FAI and AFAI were proven to be robust in the detection of floating blooms over conditions with different aerosols, sun glints, and solar zenith angles [6], [16]. We compared the R_{rc} values and FAI or AFAI values with match-up restrictions applied to the selected matched pairs, and they had a better agreement in conditions with similar band settings. The higher variations between the FAI of MODIS or VIIRS and AFAI of GOCI contributed to the uncertainties in the derived FAC. In addition,

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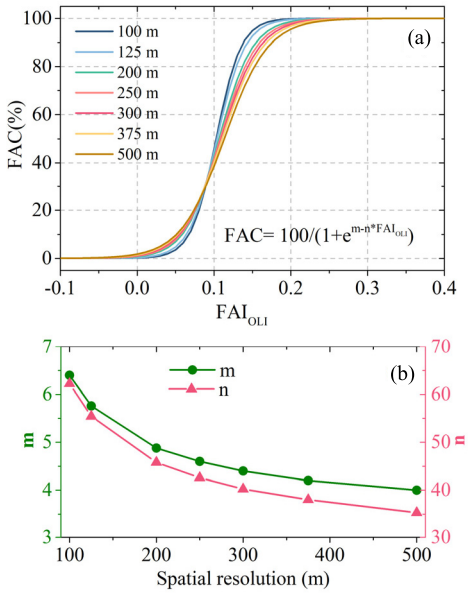


Fig. 19. (a) FAC model tuned using FAI of subsample OLI data to low spatial resolutions (100, 125, 200, 250, 300, 375, and 500 m). (b) Variations of m and n at different spatial resolution.

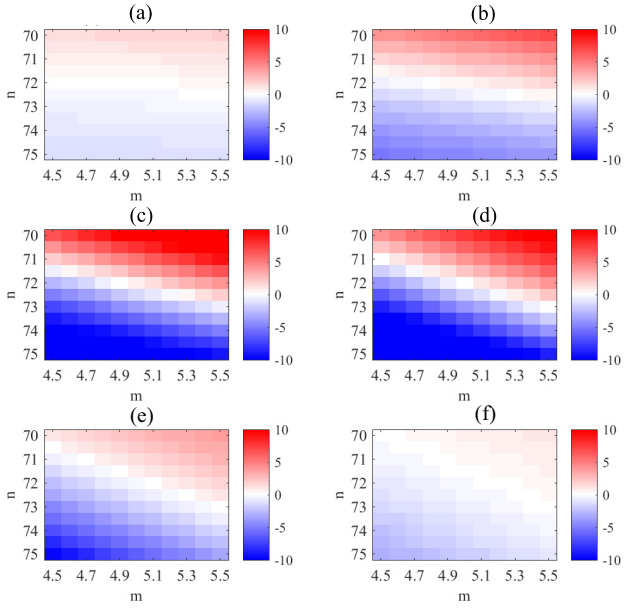


Fig. 20. Sensitivity of the FAC model [ΔFAC = FAC(m, n)-FAC(5.0, 73.6)] to varying m and n under different AFAI values of GOCI. (a) AFAI = 0.02, (b) AFAI = 0.04, (c) AFAI = 0.06, (d) AFAI = 0.08, (e) AFAI = 0.10, and (f) AFAI = 0.12.

the width of the band also contributed to the variations among different sensors. Using field- or satellite-measured high-spectral-resolution references covering floating blooms could assess the relationship between FAI and AFAI in different sensors more accurately, which should be further studied to analyze the influence of band setting and spectral resolution.

The uncertainties also come from variations in the spatial resolution of multisensor satellite data. In order to analyze the influence of spatial resolution on the FAC model, the 25 m OLI R_{rc} data were subsampled to low spatial resolution (100, 125, 200, 250, 300, 375, 500 m), and then the FAI of each spatial resolution was derived following (1). The bloom percent at each spatial resolution and the geometrically averaged FAI were modeled following the *S* curve function to determine the m and n of each spatial resolution (see Fig. 19). This illustrated that the variation of m and n decreased with increasing spatial resolution, especially from 250 to 500 m. In order to analyze the sensitivity of the FAC model, ΔFAC[=FAC(m, n)-FAC(5.0, 73.6)] was calculated to show the influence of m and n on the FAC model of GOCI with varying AFAI from 0.02 to 0.12 (see Fig. 20). This indicated that the FAC model of GOCI was insensitive in waters with low AFAI (0.02) and high AFAI (0.12), with most ΔFAC values falling between −10% and 10%.

Considering the rapid diurnal change in floating blooms, we used the GOCI data as the reference for consistency evaluation among different sensors. In fact, GOCI data showed that the variation of EBA within 1 h can reach up to 30%, therefore the match-up rules within 1 h also had uncertainties. The maximum gradient method was used to monitor the bloom area in Landsat data, that is to say, the mixture of floating blooms and water was missed in the Landsat pixel. Previous studies have proven that the airborne-identified small bloom patches (in the order of a meter or sub-meter) can be missed in Landsat data [54] or GaoFen-1 (GF-1, 16 m resolution) [39]. In this study, the FAC model is mainly used to estimate the bloom density of middle and low spatial resolution, thus the pixel un-mixing of OLI was not taken into consideration.

C. Implications for Lake Monitoring Using Multisensor Satellite Data

Rapid increases in economic activity, population, and nutrient loads from wastewater have accelerated the occurrence of algal blooms in Lake Taihu and Lake Chaohu since the 1980s. In addition, both are shallow lakes with an average residence time greater than six months, which leads to the accumulation of cyanobacteria in these eutrophic lakes [55]. Since 2007, several lake management measures, such as point source pollution, nonpoint source pollution, and hydrologic management, have been taken to restore Lake Taihu and Lake Chaohu [42], [56]. However, intensive restoration efforts have not achieved the desired results. The main reason is that nutrient loads from wastewater and nonpoint source pollution are still high under the high increase in economics [42], [57]. In addition, climate warming, spring temperatures, and wind speed reduction are important factors in lake eutrophication [15], [55]. Climatic changes have provided an additional catalyst for the expansion of floating blooms. In particular, rising spring temperatures have accelerated the onset and proliferation of Microcystis spp., which accounts for blooms in the middle and lower reaches of the Yangtze River basin [55].

Monitoring floating blooms using multisensor data can not only capture more bloom dynamics under different conditions but also provide comparable long-time series data with satellite

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development. Importantly, the time resolution should not be overlooked in analyzing long-term FAC data. By adding GOCI data, more bloom events were observed than when using MODIS data only. From the diurnal analysis of FAC, we can see that the maximum bloom area occurred at different times within a day. Therefore, using MODIS (AQUA), which overpasses at approximately 2 P.M. at Lake Taihu and Lake Chaohu, the diurnal variations would be overlooked. Even though GOCI expired in March 2021, GOCI-II was launched in February 2020, with a spatial resolution of 250 m and 13 wavelengths, giving it the potential to take over the role of GOCI in this region. With the performance degradation of MODIS, VIIRS and OLCI can succeed in monitoring daily floating blooms using the proposed FAC model, and could provide comparable results with MODIS and Landsat data. We could also build a longer history of bloom variations from the 1980s; however, due to the time-frequency of Landsat data, it is not recommended to analyze the yearly or monthly dynamics of algal blooms using Landsat data only, so the ten-year period was used for monitoring the blooms of lakes across the globe [12]. Moreover, for small lakes, aquatic vegetation and adjacent effects may influence the detection of floating blooms; thus, more satellite data with high spatial resolution, such as OLI, MSI, and HY-1C/D CZI (coastal zone instrument), should be combined to fill the gap in the time resolution.

V. Conclusion

The consistency of R_{rc} and floating bloom detection index of multisensor satellite data with the low-middle spatial resolution was assessed. Match-up pairs in 15 virtual stations of Lake Taihu and Lake Chaohu showed that high agreement in FAI or AFAI was achieved, even though R_{rc} values had a relatively lower agreement due to the spectral band settings. To address the issue of overestimation of floating bloom area using pure pixel assumption, a new logistic decomposition (S-curve) method was proposed for detecting fractional floating algal cover from satellite images with MODIS, VIIRS, GOCI, and OLCI. EBA derived from the proposed model agreed with the bloom area of OLI. Then, the FAC and EBA of multisensor data over Lake Taihu and Lake Chaohu from 2002 to 2020 were derived, which could provide a sustainable FAC dataset with the development of satellite sensors. Further studies in fractional floating algal cover monitoring are suggested to form a robust rule which is not limited to the spatial or spectral difference among different satellite sensors, and explore active remote sensing, such as Synthetic Aperture Radar (SAR), in floating bloom detection, to broaden the floating bloom information in cloudy conditions.

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